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RESEARCH-ARTICLE

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In the last two decades, autism research in computing has experienced tremendous growth, highlighting the potential to improve the experience of autistic individuals. Our research offers a detailed understanding of the landscape of autism computing research in the ACM Digital Library by conducting a literature review (N=533). Our analysis reveals that autism computing research lacks inclusiveness in several important demographic dimensions, including gender identity, age, and culture. The developed technology is largely focused on diagnosis (e.g., autism detection) and addressing the needs of autistic children (e.g., improving social communication) while ignoring the challenges of autistic elderly and adults (e.g., independent living and employment). Also, autism computing research in the ACM Digital Library is concentrated in North America and Europe, with no or little representation from Africa, South America, and Oceania. We identify areas that deserve attention and provide actionable guidelines to enhance the inclusiveness and generalizability of autism research in computing.

CCS Concepts: • **Human-centered computing** → **Accessibility technologies**.

Additional Key Words and Phrases: Autism, Computer Science, ACM Digital Library

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1 INTRODUCTION

Computing research on autism refers to the use of computers or similar technology (e.g., mobile phones, robots, smartwatches) to assist autistic individuals. This thread also includes research that aims to understand autistic individuals' experiences in everyday life and with technology, often to envision better autism support technology. Technology is thought to be well-suited for autistic individuals and autistic stakeholders (e.g., practitioners such

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as psychiatrists, therapists, special educators, caregivers) for multiple reasons. First, technology can capture real-time data valuable for retrospective analysis and design and delivery of just-in-time interventions [3]. Second, well-designed technology is accurate and objective, increasing the potential reliability of diagnosis and intervention design [9]. Finally, technology has been identified as a major interest for autistic respondents [60] with various technologies already commonly recommended as daily stress relief aids for them [33, 45].

Computing research on autism progresses through natural evolution and follows technology trends. As new technologies emerge and become popular, autism computing research tends to feature these new technologies more, irrespective of target users' needs and their application to the field. Existing research suggests that there is a mismatch between the technology needs of autistic individuals and what is available or designed [42].

Specific imbalances in autism computing research are worth noting. One such example is the gender ratio of diagnosis — autism is diagnosed more often in males than in females and any other gender identities. While there may be biological factors contributing to this imbalance, research suggests that this gender imbalance is partially caused by misdiagnosis due to the use of diagnosis scales that may be inappropriate for non-males [13].

The increasing use of data-driven technology (e.g., machine learning) to aid in diagnosis can potentially perpetuate this imbalance. The potential under-representation of autistic females and non-binary autistic individuals' data in public or collected datasets used for training machine learning models can lead to models that will contribute to the misdiagnosis of these autistic populations. Another important consideration is the location where research is conducted, as culture often drives technology design, adaptation, and appropriation [15, 56]. For example, certain technologies (e.g., robots) may be inaccessible to underdeveloped countries due to geographic, political, or economic concerns. Furthermore, the region's culture may also contribute to its approach to autism and consequently impact its research strategy and quantity.

Acknowledging the tremendous potential of autism research in computing and the need for careful research design, we investigate the landscape of autism research in computing to identify areas that work and areas that warrant more research attention. Our review adopts the neurodiversity lens, which conceptualizes autism as a natural variation in neurological functioning. This lens emphasizes individual autonomy, lived experiences, and advocates for environmental accommodations rather than conformity. We aim to answer the following research questions with reference to autism computing research published in the ACM Digital Library:

- How has autism computing research evolved over time?
- What problems (e.g., diagnosis, intervention) are researchers trying to address? Which research areas are highly represented and which are under-explored?
- Does autism and computing research align with the needs of autistic individuals?
- Are current autism-computing technologies inclusive and suitable to represent the diverse range of autistic people across gender, age, cultural, and neurological differences?
- What language have researchers chosen when discussing autism? Has this language shifted over time, and how does it impact the designed technology and autistic individuals?

To answer our research questions, we conducted a keyword-based search on all peer-reviewed autism and computing articles available in the ACM Digital Library, which resulted in 533 eligible articles. We conducted a thematic analysis on a dataset consisting of the articles' titles, abstracts, methodology, metadata (e.g., locations of authors' institutions, publication date), and keywords. Among other findings, our analysis signifies a critical need for a more inclusive research design in several important demographic dimensions (gender identity, age, and culture). In addition, our findings reveal a misalignment between available technology and the needs of autistic individuals and the autism community. By identifying areas that deserve more attention, we propose guidelines for designing next-generation autism support technology. We believe findings reported in this paper will help advance the field of wearable, mobile, and ubiquitous computing systems that are increasingly more utilized to assist the autism community.

1.1 Positionality

Our research team consists of a combination of autistic, neurodivergent, and allied perspectives. Together, our team also represents various cultural backgrounds and gender identities. Our research aligns most closely with the social model of disability [19, 47], viewing autism as a neurodiversity. We recognize the multi-faceted lived experiences within the autism community, including those who view autism as an inherently disabling condition. As a team, we strive to uplift autistic strengths [59] centering Universal Design for Learning (UDL) principles [36] and the appreciation of autistic cognitive differences [1].

Furthermore, we believe that researchers have a responsibility to choose their language carefully to avoid perpetuating negative sentiment, which is often present in autism computing research due to historical influences. We believe that researchers need to be better aware of autism as a domain and autistic individuals' preferences and avoid using ableist language, which may contribute to better acceptance of autism in society, a sentiment also echoed in [6].

2 RELATED WORKS

As noted in prior literature, technologies for autism in HCI have primarily focused on young boys and often reflected a Western and medical-model perspective [38]. Gaps have been identified in research representing the African continent, adult and elderly demographics, and the development of culturally informed technologies that consider diverse religions, family sizes, cultural norms, and languages [8, 39, 51]. Multiple reviews have examined various age groups of autistic individuals, exploring their agency in research and noting the influence of neurotypical clinicians, caregivers, educators, and researchers in shaping study topics [38, 54]. Autism computing research has faced criticism for its disconnect from critical disability studies and autistic adults, often prompting defensiveness from neurotypical researchers when questioned about potential harms [18].

Researchers noted a disproportionate focus on autism compared to other disabilities in assistive technology research [4]. Existing research have explored specific technologies, including wearables, virtual reality (VR), machine learning, robots, and games [44, 49, 51, 52, 55]. In addition, these research pointed out the need for alignment of research focus and autistic needs, identifying the potential for dehumanization. For instance, humanoid robot mentors providing socialization guidance to human children were seen to further harmful narratives, posing robots as more "human" and capable of social interaction than autistic individuals [49]. In game design, one concern was the lack of self-determination in games for children; gamifying tasks instead of creating an implicitly fun "game" [55]. Additionally, recent research on autism in computing highlighted a need for researchers to come up with robust and standard evaluation metrics for the envisioned and developed technology, explaining that assistive technologies should not introduce barriers [44]. They suggested the slow introduction of new technology into target users' lives, simple-to-use technologies that are invisible and do not "out" the user as disabled, and being critical of technology use as an "evaluation tool," wherein the user's progress is tracked, as technology as a feedback mechanism can be perceived negatively [51]. Assistive technologies should also be easy to set up and account for misuse [51].

Our analysis tracked language, determining the prevalence of person-first, identity-first, mixed, and scientific language over time. Tracking autism categorization also revealed the prevalence of research on specific demographics, such as speaking autistic individuals, Asperger's, and varying support needs. Our findings highlight the insufficient inclusion of autistic individuals with higher support needs, such as ASD level 3 and non-speaking individuals. Few studies have quantitatively tracked these domains of autism-computing research. Like an "audit" of autism computing research, we were able to analyze the field holistically (within the ACM publication venue), highlighting underrepresented technologies, common research motivations, and over-researched technologies. We also considered non-binary and transgender representation and the lack of elderly representation, which are under-researched. Our paper is timely, considering the growth of AI in the past five years, adding to the

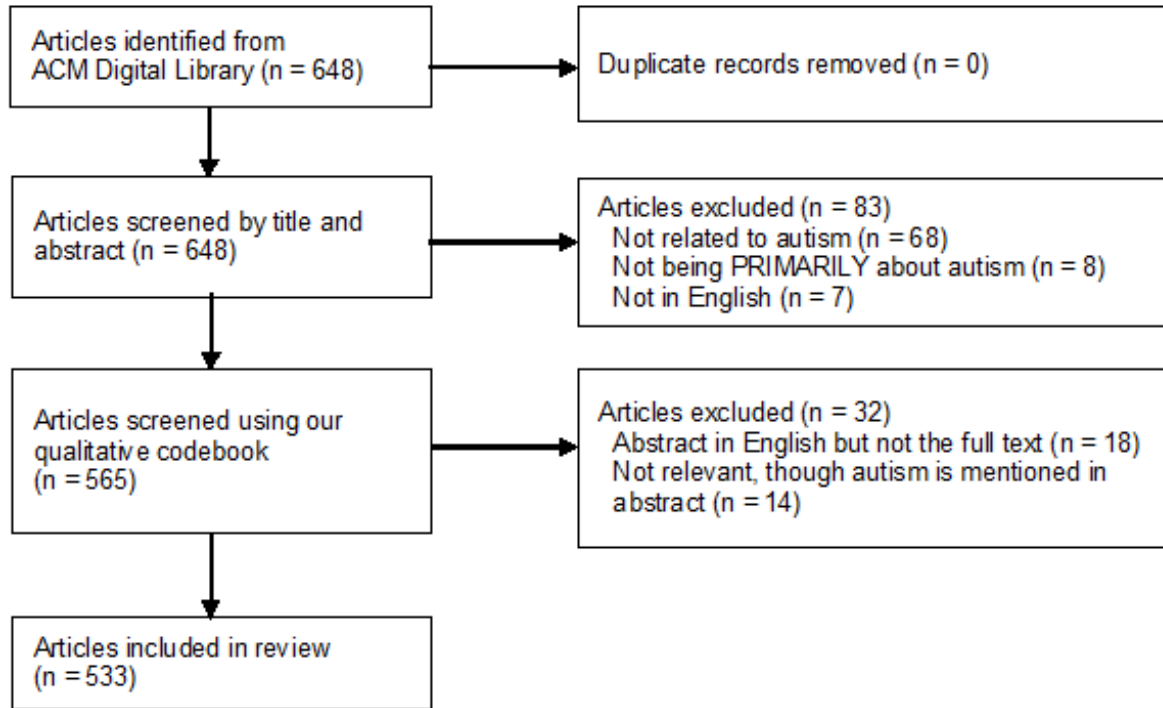


Fig. 1. Paper Selection Process.

discussion surrounding LLMs, data privacy, ethical uses, and biased data. Related works tended to focus on a specific area of research, such as a specific age group or specific technology, often minimizing their abilities to review the state of autism computing research holistically.

3 METHODOLOGY

3.1 Search Process

To collect the articles, we adhered to the basic PRISMA guidelines (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) [41]. Figure 1 shows our paper selection process. On July 12, 2024, we conducted a keyword-based search in the ACM Digital Library. We used the following query:

("autis* OR asperger* OR pervasive development* disorder* OR ASD OR PDD-NOS")

We limited our search query to anywhere in the abstract or document title of peer-reviewed research articles published in journals, magazines, and proceedings only. This resulted in 648 articles (journals: 68, conferences: 578, and magazines: 2).

3.2 Paper Selection Process

We uploaded all articles to the Mendeley Reference Manager and sorted them according to relevance based on the following exclusion criteria: first, 68 papers were excluded for not being related to autism (such as 'ASD' in the search query matching with Adjustable Speed Drives or Agile Software Development), second, eight papers

were excluded for not being primarily about autism (such as mentioning autism in the abstract but focused primarily on ADHD or depression), and finally, seven papers were excluded for not being written in English. Overall, we excluded 83 articles (68+8+7), resulting in a corpus of 565 (658-83) relevant articles. We focused on peer-reviewed research articles that were published in journals, proceedings, and magazines within the ACM Digital Library. We excluded extended abstracts but did not apply any additional exclusion criteria based on paper length or format. While we recognize that the ACM Digital Library can also contain extended abstracts or other non-traditional formats such as book reviews, our keyword search and filtering process prioritizes peer-reviewed research contributions. As such, the final corpus predominantly consisted of full papers, and we did not remove shorter submissions unless they were found to be unrelated, non-English, or non-autism-focused.

We uploaded 565 articles to ATLAS.ti Qualitative Coding Software (web version) for further analysis. We excluded 32 additional articles from this corpus for two reasons. First, 18 articles had abstracts written in English, but the full texts were written in a different language. Second, reviewing the methodology during this phase revealed that 14 selected articles were irrelevant despite having autism in the abstract. Excluding these 32 articles (18+14) resulted in a corpus of 533 articles (565-32). We report our findings in Section 4 based on these 533 articles.

3.3 Data Analysis Process

We adopted a mixed-method approach to analyze our final corpus (N=533), utilizing a deductive qualitative content analysis and a quantitative frequency analysis. Based on our prior experience with autism computing research, we collectively designed a codebook containing 11 predefined themes, as mentioned below.

- **Problem:** This theme refers to the proposed outcome of a paper.
- **Technology:** This theme refers to the type of technology the researchers proposed, designed, or explored.
- **Challenge:** This theme refers to the paper's identified research question, focusing on the goal driving this research. For example, the codes are created to describe the characteristics of autism or the needs the research is trying to address through the development or use of specific technology. Some example codes include social skills, emotional skills, executive functioning skills, motor skills, and sensory skills.
- **Participant Number:** This theme represents the number of autistic participants researchers recruited for their research. We used the following codes to represent the participant number: small ($N \leq 10$), medium ($10 < N \leq 30$), and large ($N > 30$).
- **Participant Gender:** This theme refers to the gender of the autistic participants researchers recruited for their research. Although some researchers included neurotypical participants and parents in their studies, we excluded their gender information when analyzing this theme. We used the following codes to represent participant gender: balanced, primarily male, primarily female, only male, and only female. We used this ratio as a split between balanced and non-balanced gender: $0.66667 \leq \frac{\text{male}}{\text{female} + \text{other}} \leq 1.5$. In addition, during coding, we marked all instances of researchers mentioning transgender or non-binary participants.
- **Participant Age:** This theme refers to the age of the autistic participants researchers recruited for their research. We categorized participant age into the following categories: child ($1 \leq \text{age} < 12$), adolescent ($12 \leq \text{age} < 18$), young adult ($18 \leq \text{age} < 30$), and adult ($\text{age} \geq 30$). It is important to point out that none of the papers included elderly autistic participants, making it futile to add a code for elderly autistic participants.
- **Language:** This theme refers to the language the researchers used in their articles to represent their autistic participants. We used the following codes to present such language: identity-first language (e.g., autistic individuals), person-first language (e.g., individuals with autism), and mixed language (e.g., both autistic individuals and individuals with autism were used in the same article). There are articles where the authors didn't explicitly mention the language (e.g., ASD characteristics). We coded those as missing the language dimension we were investigating.

Table 1. Sample themes and example codes.

Theme	Number of unique codes	Example Codes
Problem	12	diagnosis, intervention
Technology	130	machine learning, social media
Challenge	60	social, eyecontact
Participant number	4	small, medium, large, missing
Participant Gender	9	transgender, primarily female
Participant Age	5	child, adolescent, youngadult, adult, missing
Language	4	person-first, identity-first, mixed, missing
Level	17	highfunctioning, aspergers

- **Level:** This theme refers to the commonly identified areas of categorization within autism. It is important to note that autism is a spectrum, with each autistic individual having varied experiences among core domains: social, motor control, executive functioning, sensory experiences, and cognitive abilities. Throughout history, researchers have attempted to categorize the diverse experiences of autistic individuals, utilizing a series of labels often closely following the diagnostic manual criteria available and in use during the research period (e.g., DSM-5). We used these labels, as mentioned by the researchers, as our codes for this theme, such as high functioning, mild functioning, low functioning, behaviorally unstable, cognitive impairment, verbal, limited verbal, non-verbal, level 1, level 2, level 3, Asperger, mild, moderate, severe, and IQ.
- **Location:** This theme refers to the countries in which the authors' institutions are located. We recorded it as one when all researchers belonged to institutions from a single country or multiple when researchers from institutions located in different countries collaborated on the publication. For example, a research article including authors' institutions from the United States and China was coded as multiple, with the United States and China recognized as the location of research.
- **Year:** This theme refers to the publication year of the article.
- **Keywords:** This theme refers to the keywords provided by the authors in the article.

Table 1 provides a sample of the themes and the number of codes associated with these themes, with examples.

Three members of our research team independently coded the articles for the first eight themes (problem, technology, challenge, participant number, participant gender, participant age, language, and level). We created codes by carefully reading the abstract and methodology section of these 533 articles. Each article was coded by one of our research team members and reviewed by a second member. All questions regarding specific code assignments were resolved during our regular discussion sessions with all research team members. This ensured coder agreement and consistency of code application (such as the differences between virtual, augmented, and mixed reality). The participant number and language codes were exclusive, meaning only one code per document was used. In contrast, the others are non-exclusive, allowing multiple codes to be assigned. For instance, if different technologies were used or different age groups were recruited, we assigned codes for every relevant category. We created custom Python scripts to collect the remaining three themes (location, keywords, and year) from the metadata provided by the ACM Digital Library.

4 FINDINGS

This section reports our findings based on the themes listed in the Methodology section. As mentioned earlier, we collected all eligible articles on autism from the ACM Digital Library, and the earliest article was published in 1971. The only other article published before 2007 was from 1998. Figure 2 presents a timeline visualization

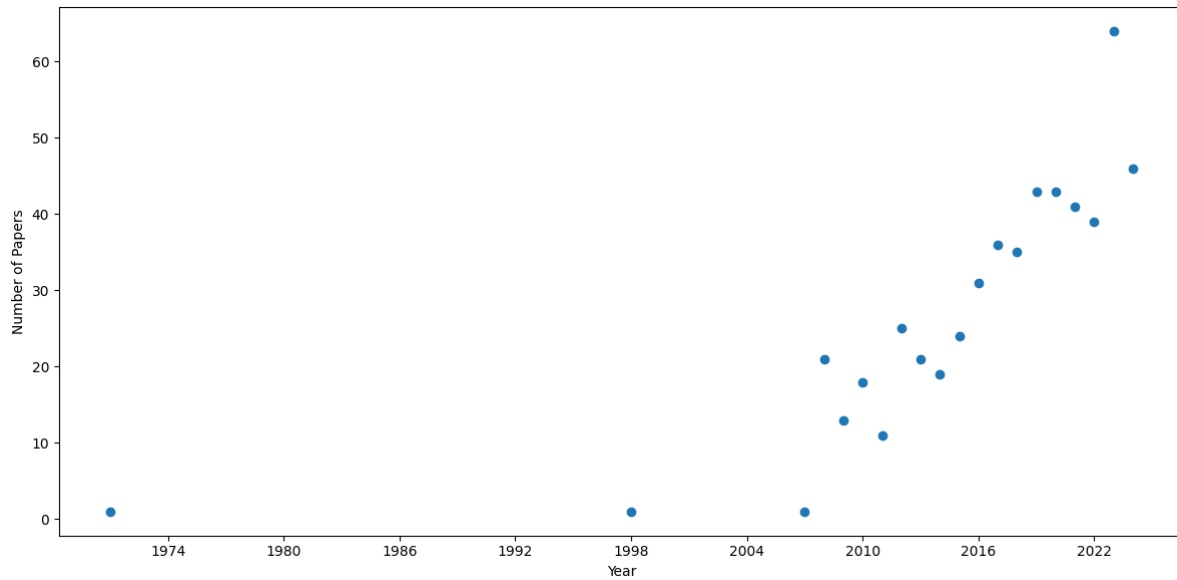


Fig. 2. Papers published over time.

of all articles published, where the x-axis represents time and the y-axis represents the total number of articles published in that year. Figure 2 highlights that since 2007, this thread of research has gained a lot of attention from researchers. To make the visualizations more intuitive by excluding the large time gap between 1971 and 1998 and 1998 and 2007, all the remaining visualizations show data starting from 2007. It is important to note that while the first data shown is from 2007, as the x-axis only included alternate years, the year 2007 is not shown as a marker in the x-axis.

4.1 Participants — Non-Inclusive in Gender, Age, and Other Demographic Dimensions

While autism impacts individuals of all ages, gender identities, and socioeconomic backgrounds, autism research in computing is often skewed to specific types of participants. In the following subsections, we report our findings regarding different demographic dimensions.

4.1.1 Gender. In 533 articles, there are 541 codes regarding the gender of the participants. In some articles, researchers recruited autistic participants from multiple gender categories. A vast majority of articles, 60.79% (324/533), didn't report gender information. The articles that reported participant gender revealed a significant gender disparity present in autism computing research. For instance, we found that 32.27% (172/533) of articles have a male-dominant gender representation (only males without any females or primarily males with a small number of females). On the contrary, 2.44% (13/533) of articles have a female-dominant gender representation (only females without any males or primarily females with a small number of males). Only 4.32% (23/533) of articles have gender-balanced representation, an aspect that warrants attention to this significant gender imbalance from researchers.

Most research reported the participants' gender using a binary classification, with a minority of papers including an "other" category. Only 1.31% of articles (7/533) utilized a non-binary gender classification. It is important to

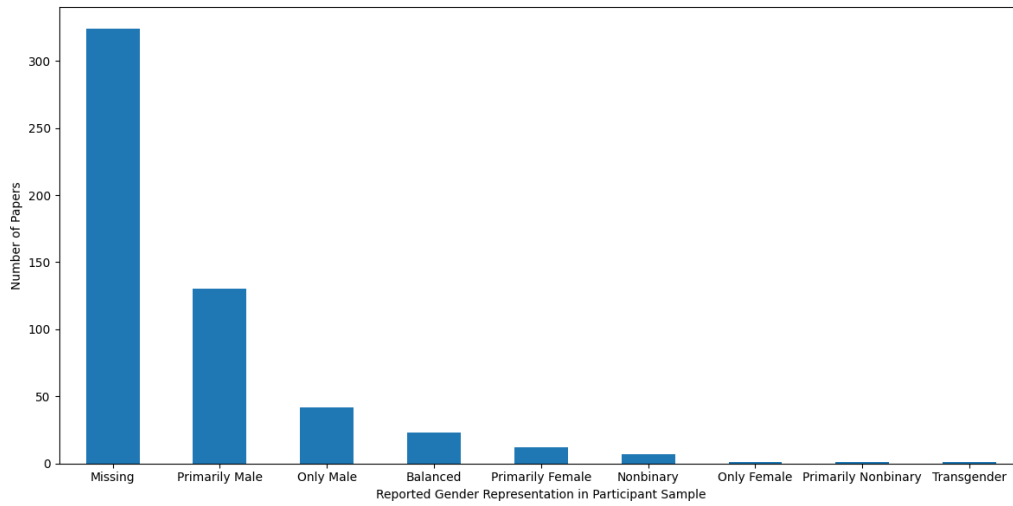


Fig. 3. Distribution of gender of autistic participants of all time.

note that articles including a non-binary classification were published after 2022, adopting an inclusive approach toward participants with non-binary gender identities.

Figure 3 shows the overall gender distribution of participants in autism computing research. We found that 60.79% of published articles did not include gender information. Figure 4 highlights the gender distribution reported in the papers over time, where the size of the dots reflects the number of samples for each category. For instance, all articles published in 2007 reported data from primarily male participants. Figure 5 highlights that until 2018, none of the articles included transgender autistic individuals. These figures highlight that even though more articles have been including gender identities, a vast number of them still do not report this critical demographic information.

4.1.2 Age. We identified 676 codes regarding the age of the participants in 533 articles. Some articles recruited autistic participants from multiple age categories. It is important to note that a significant number of articles, 23.64% (126/533), did not report the age of their participants or did not recruit any participants. For the remaining articles, we found that the age of the participants is skewed towards children. Of the articles that reported age, 57.79% (308/533) focused on children, 19.51% on adolescents (104/533), 15.76% on young adults (84/533), and only 10.13% on adults (54/533). 22.70% (121/533) of these articles had more than one age group represented among their participants. It is important to note that no article collected data from or reported findings based on elderly participants. Figure 6 shows a timeline visualization of the age distribution of the participants, highlighting the dominance of research focused on autistic children, where the x-axis shows time and the dots show the number of articles reporting the specified age group. It is interesting to note that all articles from 2007 focused on young adults.

4.1.3 Location. We analyzed the countries of the authors' institutions. 93 articles (17.45%, 93/533) have authors from institutions that are located in multiple countries. Although the authors came from 55 separate countries, most articles were written by authors from five countries, all of them considered economically developed regions: USA (39.02%, 208/533), UK (11.63%, 62/533), China (10.13%, 54/533), Italy (6.00%, 32/533), and Australia (5.63%, 30/533). Among 55 countries, 30 have representation in 5 or fewer articles. Figure 7 shows the distribution of

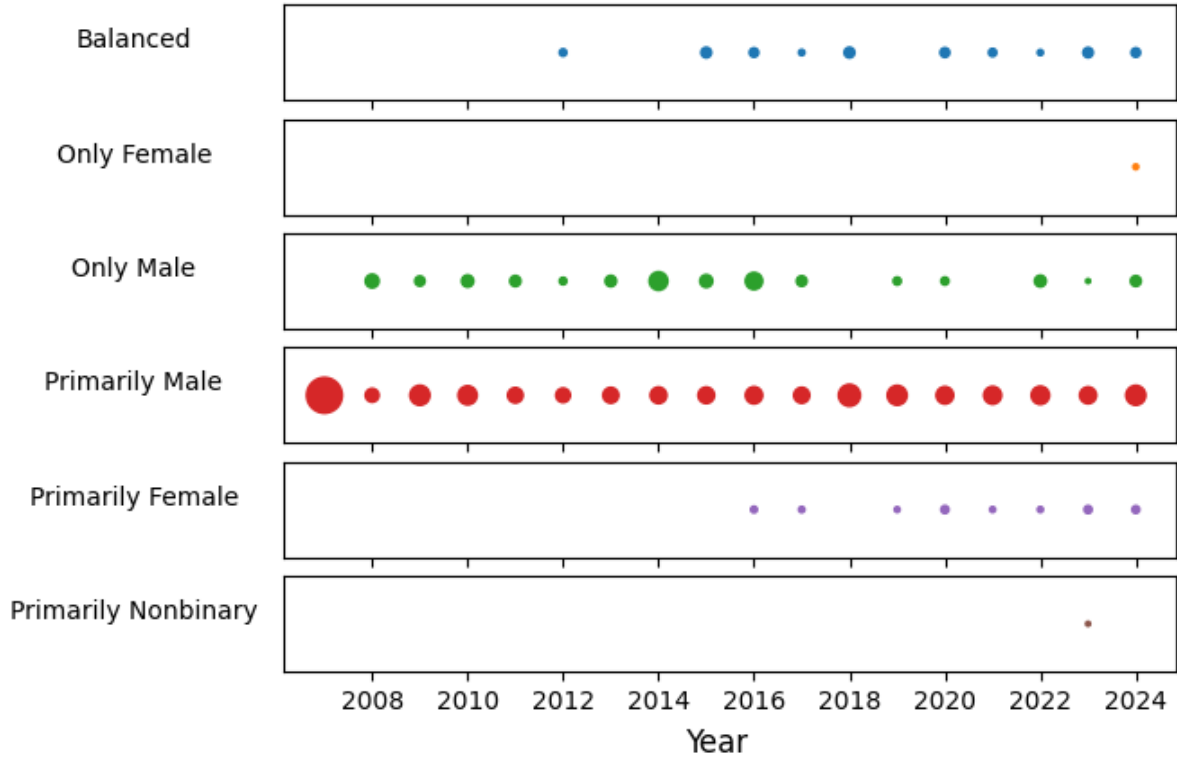


Fig. 4. Timeline distribution of gender of autistic participants (normalized).

the countries of the authors' institutions. Figure 8 uses a heatmap to show the location-based distribution of autism computing research, highlighting geographical areas where most of this research is concentrated. This figure shows that although only 13.1% of the global population lives in North America and Europe [48], they contributed the most to autism computing research in ACM Digital Library. On the contrary, Africa has 18.93% of the world's population [48]. Unfortunately, no articles on autism in computing, which were published in ACM Digital Library, were written by researchers residing in any African country.

4.1.4 Types of Autistic Participants. The type of an autistic participant refers to the theme "level" discussed in the Methodology section. From 533 articles, we determined 624 total codes and 17 unique codes regarding the types of autistic participants focused on and described by the researchers. A vast majority of the articles, 65.85% (353/533), did not specify the participants' autism characterization or did not have participants. In some articles, researchers recruited multiple types of autistic participants. It is important to note that the DSM-5 was introduced in 2013, which redefined autism as having three levels: ASD Level 1, ASD Level 2, or ASD Level 3. These levels are based on the types and intensity of support autistic individuals need in their everyday lives. Before 2013, terms such as high-functioning, mild-functioning, low-functioning, severe, moderate, and mild were prevalent. These trends in language are likely influenced by the adoption of new terms for diagnostic criteria in the DSM-5 as compared to DSM-4 or its predecessors.

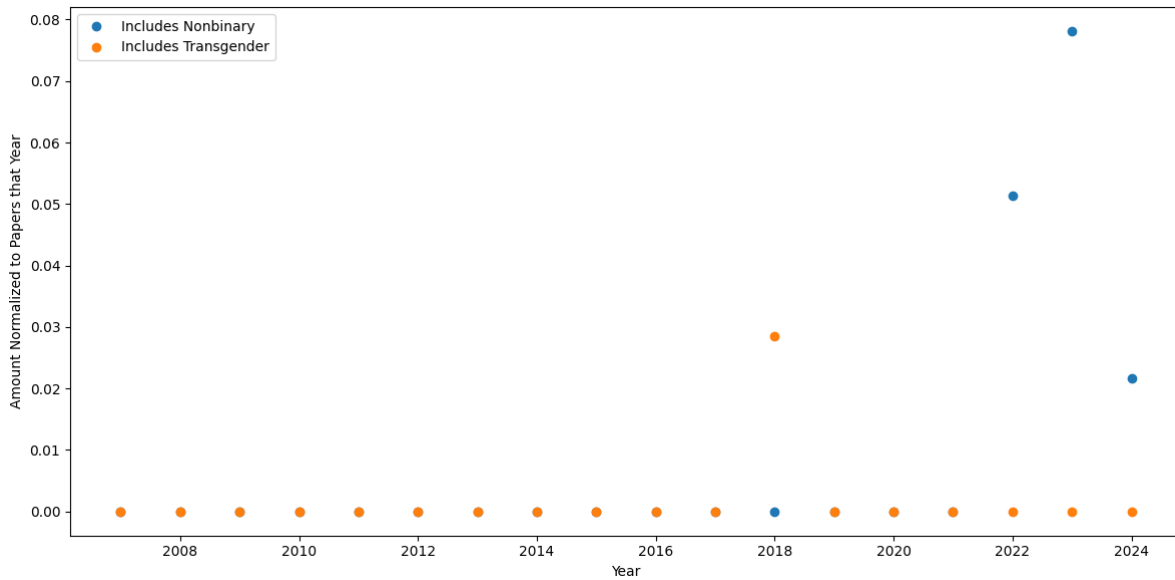


Fig. 5. Timeline of inclusion of transgender and non-binary autistic participants.

Figure 9 shows the timeline distribution of three types: Low Functioning, Mid Functioning, and High Functioning, with the x-axis showing time and the dots showing the number of papers reporting the specified terminology. Similarly, Figure 10 shows the timeline distribution of three types: Severe, Moderate, and Mild, using the same axis categories. In addition, Figure 11 shows the timeline distribution of three types: ASD Level 1, ASD Level 2, and ASD Level 3, using the same x-axis values. We observed that even after the terminologies were finalized by DSM-5 in 2013, many researchers continued to use earlier categorizations (e.g., high functioning, low functioning) even for articles published in 2024. Figure 11 also points out that the introduced terminologies started gaining popularity in 2019.

Apart from these levels, another essential distinction for autistic individuals is whether they are speaking, non-speaking, or minimally-speaking (previously referred to as verbal, non-verbal, or minimally-verbal). Figure 12 shows the distribution of articles with participants having these characteristics. The x-axis shows time, and the size of the dots shows the number of articles reporting the specified participant type. This figure shows that while researchers initially focused on speaking (verbal) autistic participants, there has been a recent increase in the recruitment of non-speaking autistic participants. In addition, apart from the categorizations mentioned above, a small number of articles, 4.50% (24/533), classified autistic participants according to their IQ scores.

4.2 Researcher Perspective, Research Focus, and Methodology Used

4.2.1 Language. The power of language is well-acknowledged in research. In the autism community, there is a long-standing debate over what should be the preferred language when referring to autistic individuals, their conditions, and associated factors. This debate is reflected in autism computing research as well. The two primary types of language used among the autism community and researchers are identity-first (e.g., autistic individuals) and person-first (e.g., individuals with autism). Of 533 articles, 59.29% (316/533) used person-first language, 22.33% (119/533) used identity-first language, 9.94% (53/533) specified neither of these two languages, and 8.44% (45/533)

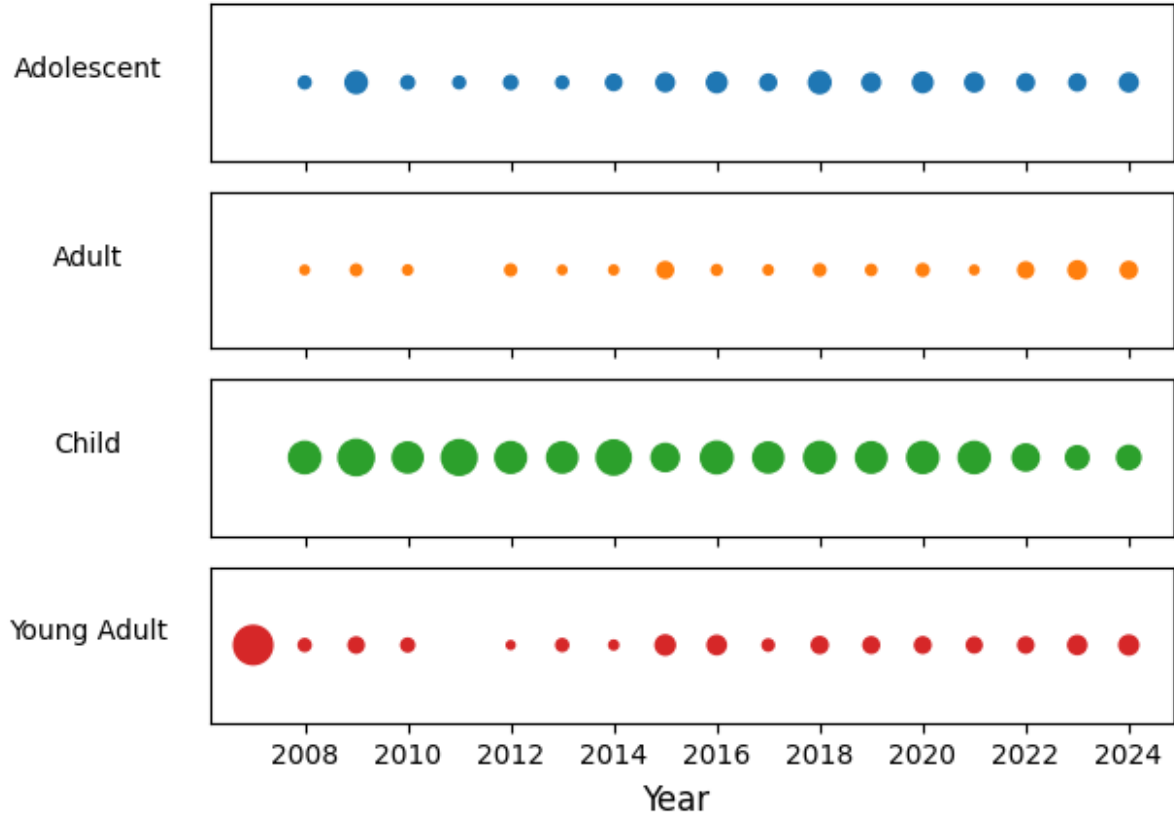


Fig. 6. Timeline of distribution of age of target user.

used a combination of the two. One example of when neither language was marked was when the researchers used language where they preferred referring to the condition directly (e.g., ASD, autism spectrum disorder). Figure 13 shows the timeline distributions of these two types of language used by the researchers, where the x-axis shows time and the size of the dots shows the number of articles using the specified language. There has been a shift in using more identity-first language in recent years, starting from 2023. Notably, in 2023, 43.75% (28/64) of articles used identity-first language, while 35.94% (23/64) used person-first language. In 2024, 45.65% (21/46) of articles used identity-first language, while 41.3% (19/46) used person-first language.

4.2.2 Research Focus. Research in the autism computing domain focused on addressing many challenges and characteristics associated with autistic individuals and autism in general. Figure 14 highlights these focus areas. It shows that the top five areas that autism computing research focused on are as follows: 63.98% articles (341/533) focused on helping autistic individuals to improve their communication skills, 27.58% (147/533) articles focused on the emotion and well-being of autistic individuals, 21.95% (117/533) articles focused on improving motor skills, 21.76% (116/533) articles focused on improving executive functioning, and 19.32% (103/533) focused on helping the cognitive processing of autistic individuals.

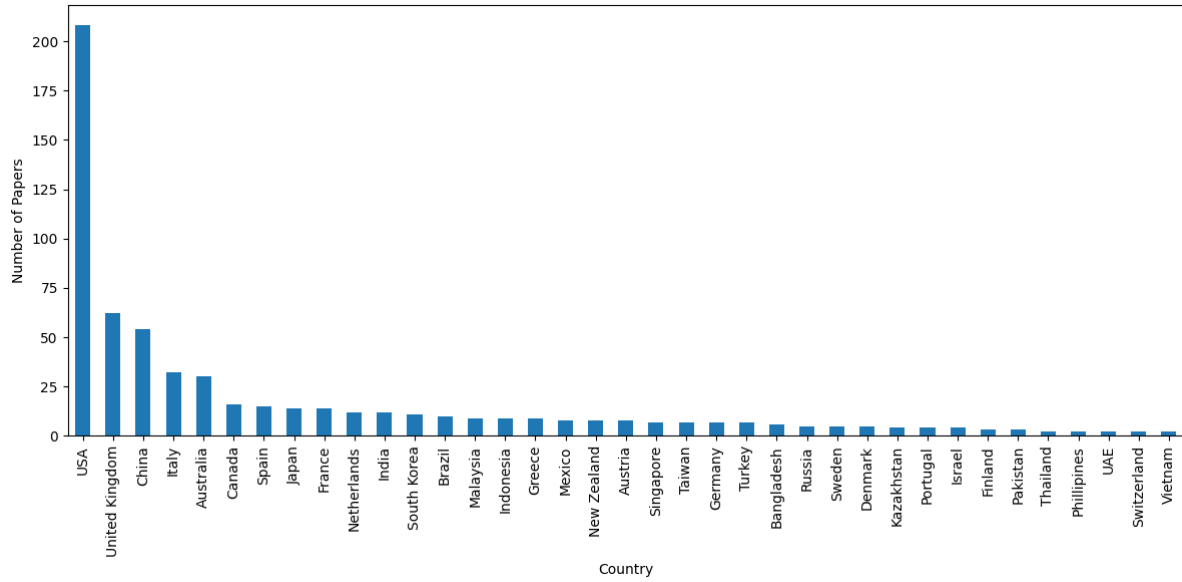


Fig. 7. Distribution of the countries of the authors' institutions.

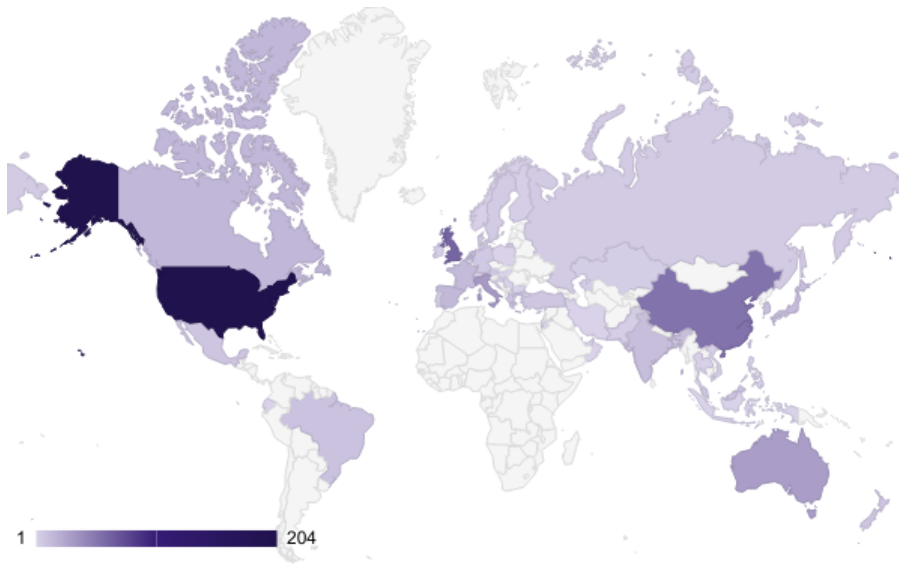


Fig. 8. Heatmap of location-based distribution of authors' institutions.

4.2.3 Nature of Data Collection. Thirty-eight articles, especially articles focused on developing machine-learning models for autism diagnosis, utilized secondary data. Of these articles, 84.21% (32/38) did not report the primary demographic details of the participants who originally contributed the data, such as their age or gender. 52.63%

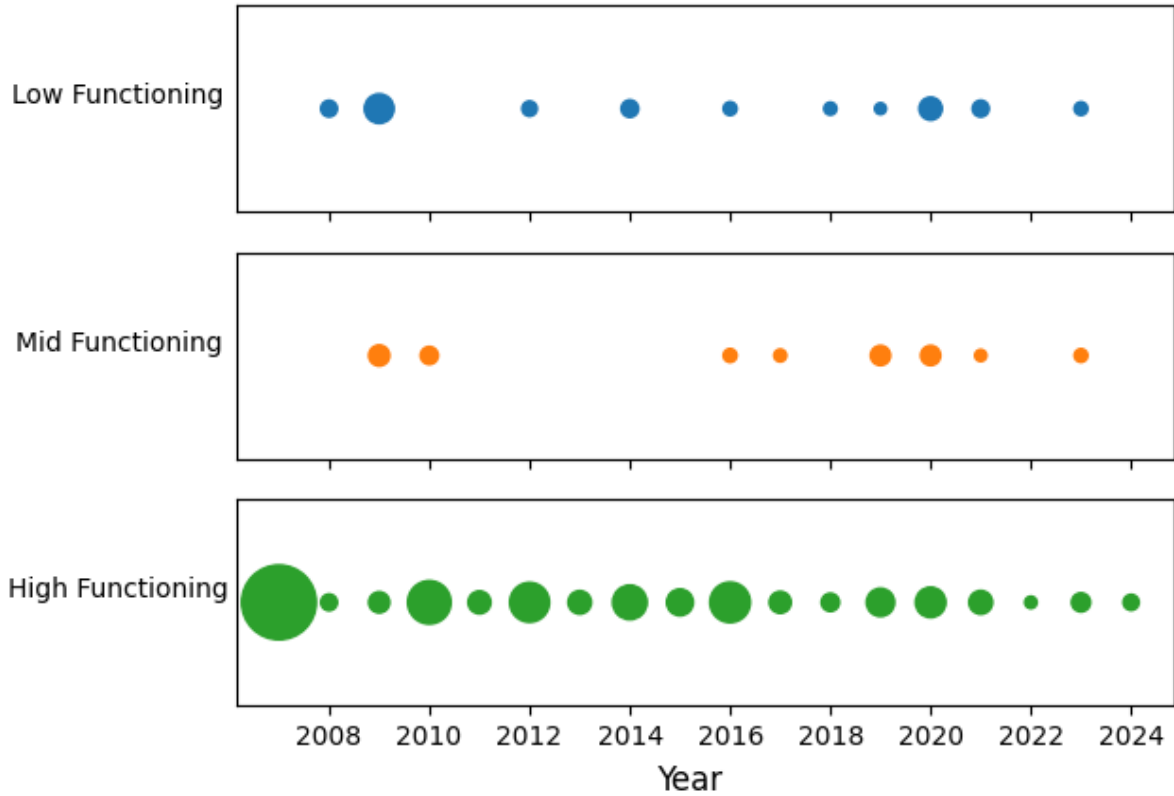


Fig. 9. Timeline of distribution of language used (low, mid, and high functioning) to refer to autistic individuals (normalized).

(20/38) articles only reported the number of participants who contributed the secondary data without reporting any additional details. In addition, 10.53% (4/38) articles did not even mention the number of participants.

4.3 Technology

This section reports various technologies designed to aid autism community.

4.3.1 Technology and Autism Areas Addressed. We found that the top five technologies used or developed by researchers in the autism computing research domain are machine learning at 12.57% (67/533), robots at 11.07% (59/533), virtual reality at 10.88% (58/533), games at 9.57% (51/533), and mobile app at 5.82% (31/533). The major problem areas these technologies are trying to solve are as follows: 43.34% (231/533) articles focused on creating intervention technology to aid autistic individuals in overcoming challenges (e.g., improving communication), 17.26% (92/533) articles focused on understanding people related to the autism domain using different techniques such as interviews and case studies, 12.20% (65/533) focused on developing diagnostic tools and models to detect autism and identify factors that can predict autism diagnosis, and 10.51% (56/533) articles focused on designing tools to analyze data, facial expressions, emotions to help researchers identify autistic and non-autistic people, their behavioral and other differences. Figure 15 maps autism areas and associated computing technology. On

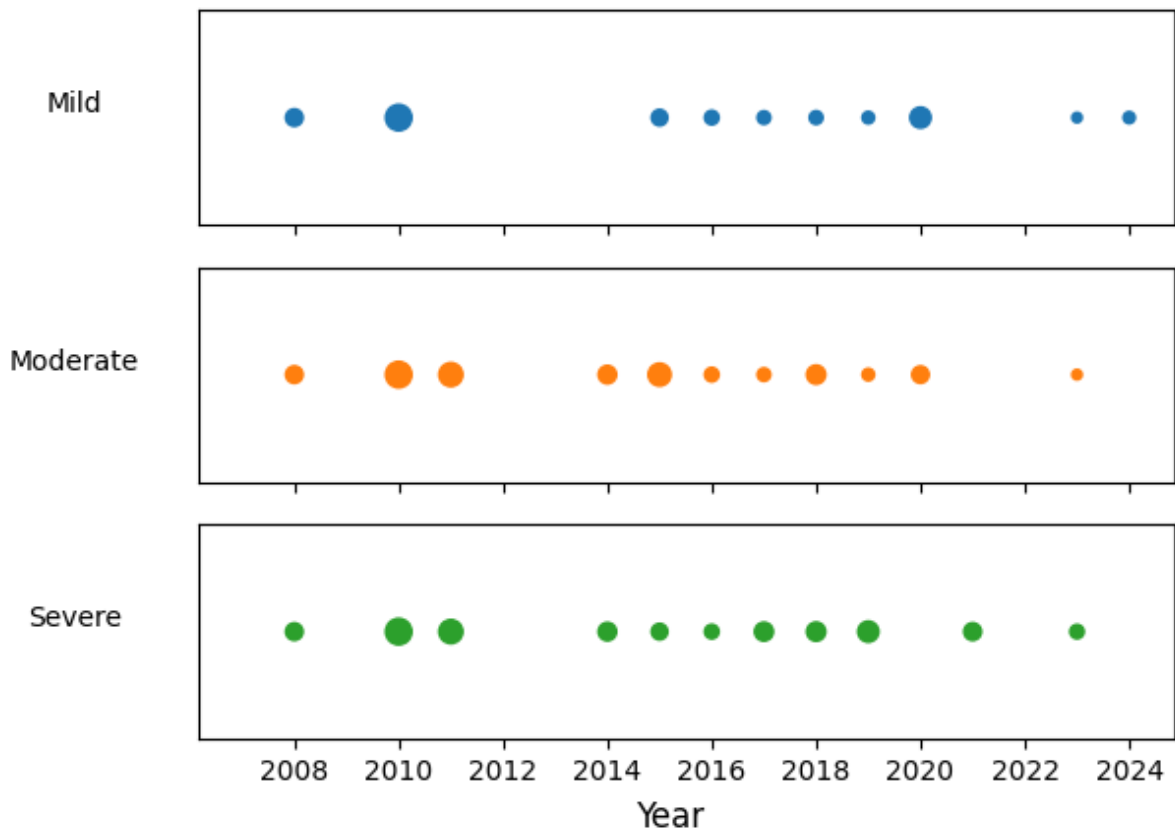


Fig. 10. Timeline of distribution of language used (severe, moderate, and mild) to refer to autistic individuals (normalized).

Table 2. Technology vs. Target Users.

	Game	Machine Learning	Mobile App	Virtual Reality	Robot
Child	35	24	17	30	45
Adolescent	13	15	4	15	7
Young Adult	6	8	2	16	2
Adult	3	3	2	4	1
Missing	9	27	11	7	10

the other hand, Table 2 shows the technology types and the target users, highlighting that most autism support technology is created for autistic children, with a small fraction created for autistic adults.

4.3.2 Machine Learning. Many diagnostic tools and technologies developed by autism computing researchers utilize machine learning and deep learning models to detect autism. 12.20% of articles (65/533) focused on detecting

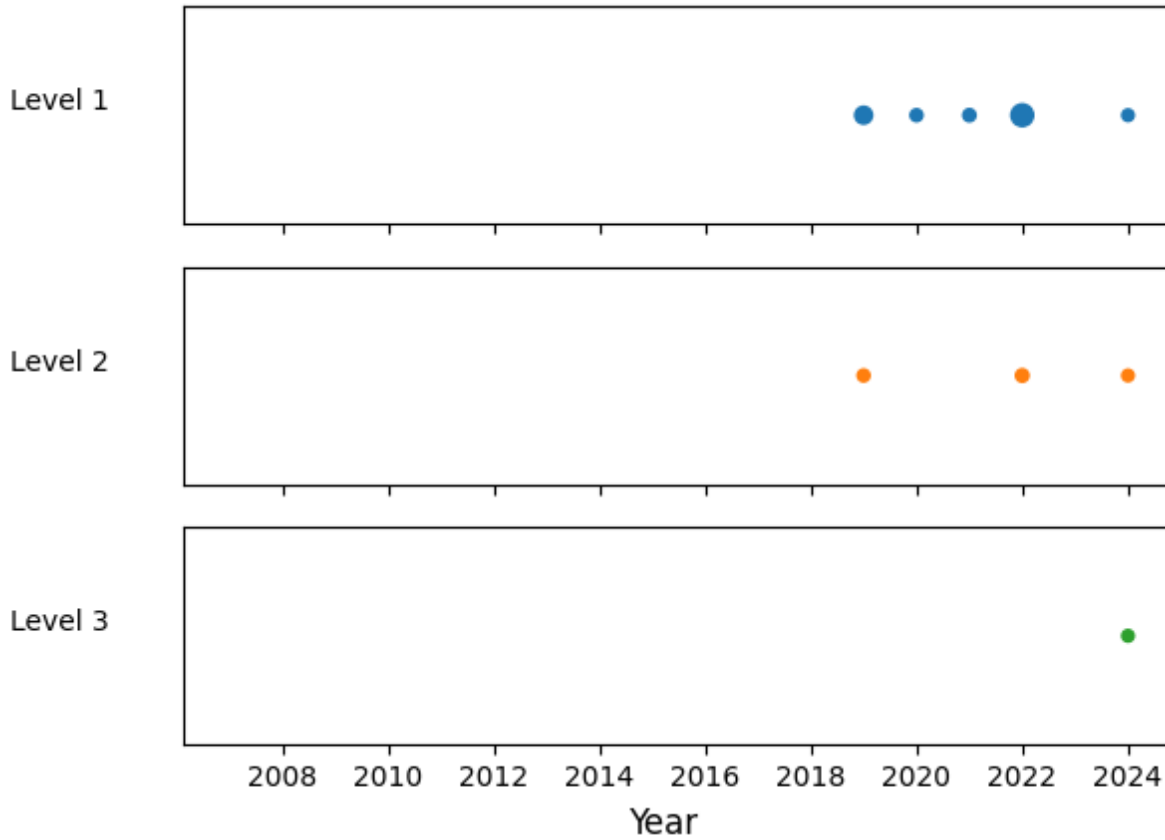


Fig. 11. Timeline of distribution of language used (level 1, 2, and 3) to refer to autistic individuals (normalized).

and classifying autism by analyzing gene data, behavioral data, and facial expressions. Another set of articles focuses on detecting eye gaze to understand social interaction and identify intervention needs and opportunities. In addition, recently, researchers have started utilizing machine learning models to predict emotions, aggression, and other behavioral traits to aid in designing evidence-based interventions. Figure 16 shows machine learning models and their focus areas. We found that 56.52% of articles that used machine learning focused on autism diagnosis, followed by analytical tools (20.29%), intervention (11.59%), and understanding (10.14%). One article was marked as focusing on both intervention and diagnosis, while the others all had one major focus area.

4.3.3 Robots as Autism Support Technology. Robots are another popular autism support technology used in multiple areas of autism. 11.10% (59/533) designed or utilized robots as autism support technology. Figure 17 shows the timeline distribution of the use of robots in autism computing research. Of the 59 research articles that used robots as a technology type, two articles focused on two problem areas (diagnosis and intervention) instead of one. Considering that, we found that 67.79% of robotics articles (40/59) focused on intervention (e.g., teaching autistic children socialization and social communication). Besides intervention, the other two major areas of robotics concentration were autism diagnosis at 8.47% (5/59) and understanding using interviews, surveys, and qualitative analysis at 8.47% (5/59).

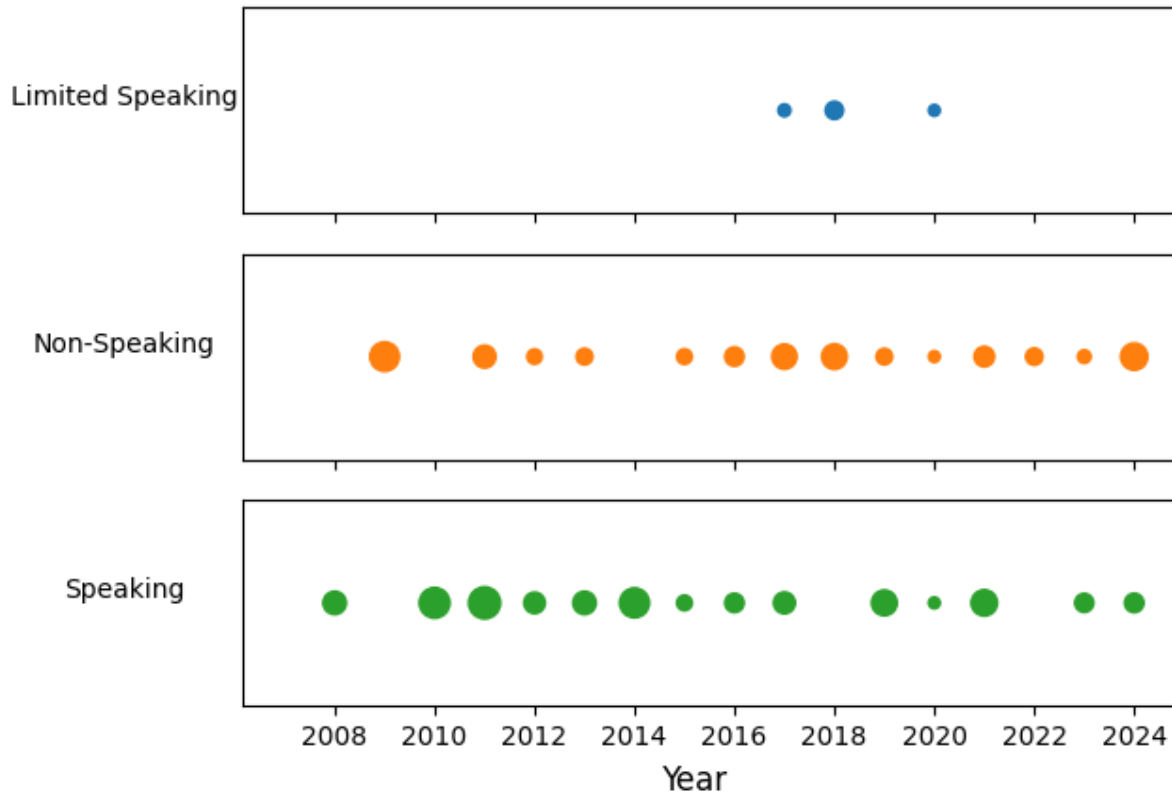


Fig. 12. Timeline of distribution of language used (speaking, non-speaking, and limited speaking) to refer to autistic individuals (normalized).

4.3.4 Mobile Apps. Another popular autism support technology is based on applications created for mobile phones and other handheld devices, such as tablets and iPads. As mentioned earlier, 5.82% (31/533) designed or utilized mobile apps as autism support technology. Like robots, articles that used mobile apps as a technology primarily focused on intervention (18/31=58.06%).

4.3.5 Emerging Technologies. In our examination of the use of technology over the years, certain technologies emerged as having only been researched and published in recent years, and not being present before. One such example is that of LLMs (large language models). Due to differences in communication and language acquisition being one of the characteristics of autism, recent improvements have made these models, especially pre-existing ones such as OpenAI's ChatGPT, appealing to researchers. In a similar trend, machine learning has seen a rapid uptick in publication count, especially in neural networks. The use of fMRI (functional magnetic resonance imaging) seems to be present only in recent publications. In addition to the above-mentioned articles, 23 articles published in the last four years presented their findings based on data analysis. Data for these articles came from genetic data sets or online social media sites.

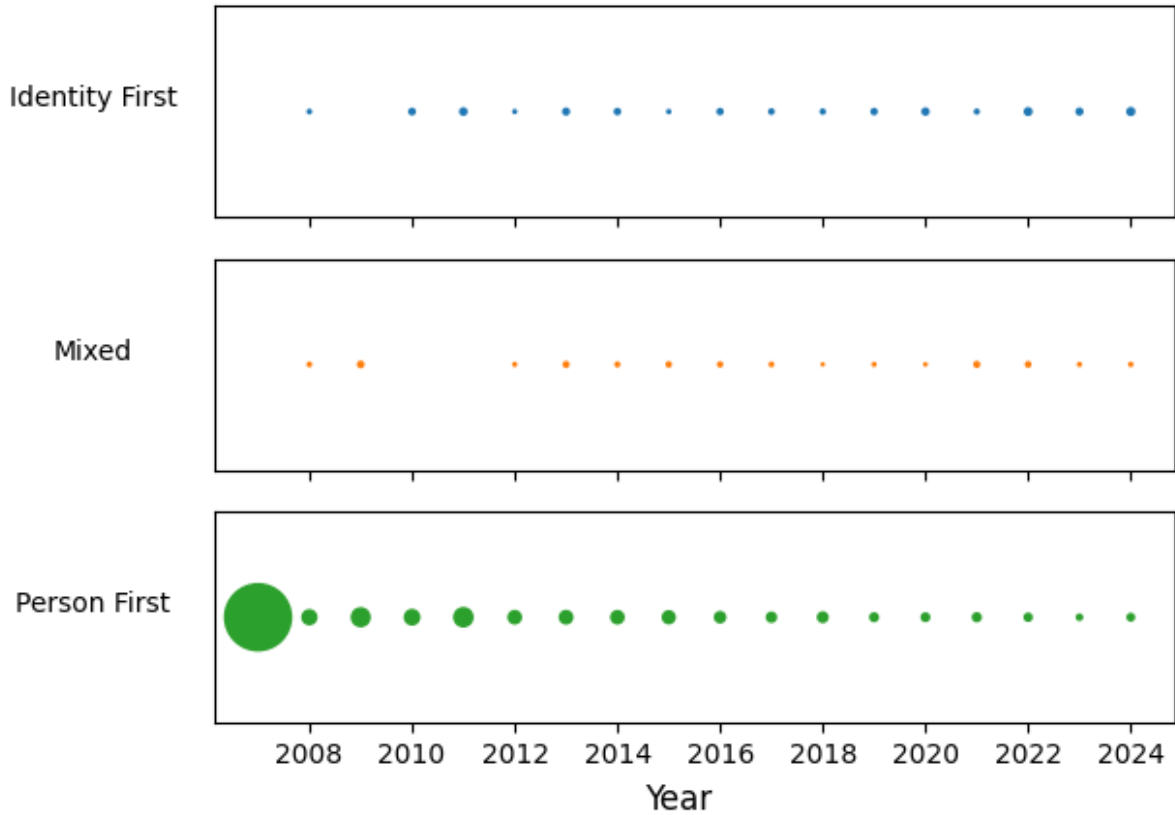


Fig. 13. Timeline of distribution of identity-first vs. person-first language (normalized).

4.4 Examining from Theoretical Lenses

In 2024, Rizvi et al. [49] reviewed 142 research articles on Human-Robot Interaction (HRI) through the lens of Critical Autism Studies (CAS), which advocated for the inclusion of autistic scholars in autism research [64]. Their analysis centered on three critical aspects of autism, namely, 1) the pathologization of autism, 2) gender and age-based essentialism, and 3) power imbalances. We acknowledge the importance of Rizvi et al.'s research framing, but wanted to extend this. As mentioned earlier, our primary goal is to provide a detailed understanding of the landscape of autism computing research in the ACM Digital Library. To achieve this goal, we adopted a mixed-method approach to analyze our final corpus (N=533), utilizing a deductive qualitative content analysis and a quantitative frequency analysis. During this process, a common theme emerged: techno-solutionism. We will examine this theme in relation to Rizvi et al.'s three theoretical lenses. Our findings extend and complement earlier work in CAS, such as that by Rizvi et al. [49], Spiel and Gerling [55], Ramirez et al. [38], and Guberman [18]. Rizvi et al.'s [49] work inspired us to reexamine our generated codes, providing an opportunity to critically analyze the codes using a power-informed theoretical lens. We employed an integrated thematic approach and analyzed our data using a CAS-informed interpretive lens.

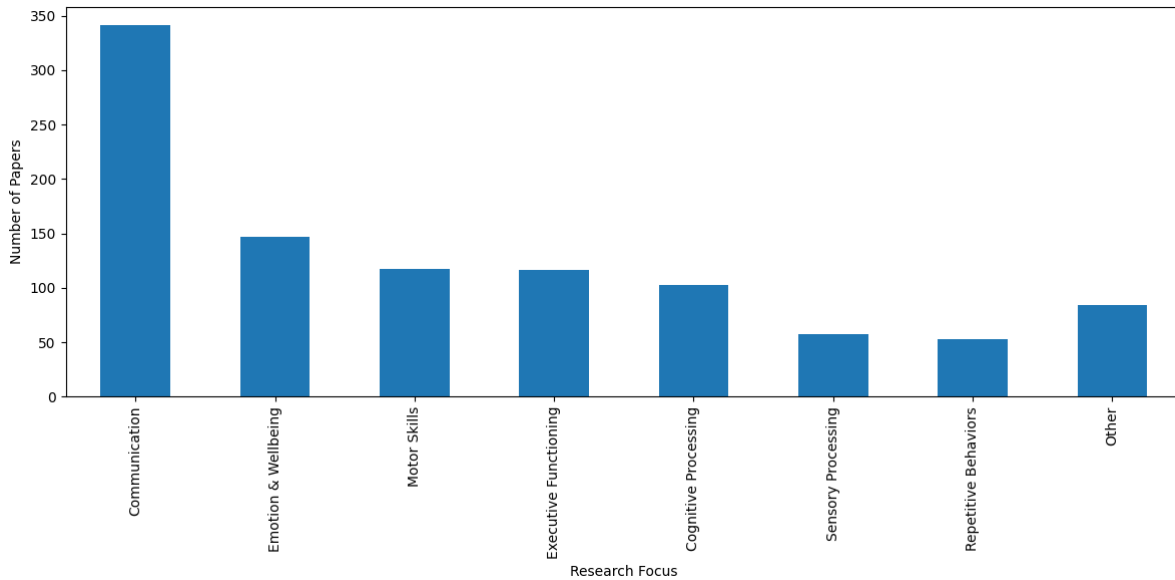


Fig. 14. Focus areas of autism computing research (non-exclusive).

4.4.1 Techno-solutionism. Techno-solutionism is the reduction of complex social issues to a problem in computation or engineering [16]. Critics of techno-solutionism argue that optimistic technological “fixes” often overlook the root cause of social issues, resulting in the reinforcement of systemic inequalities, diversion of attention from social and institutional reform, or failure in real life due to issues such as financial accessibility and user perspectives. Additionally, tech fixes risk encoding researcher assumptions and bias into technologies, and sometimes replacing the potential for human connection (ex., an AI tutor replacing instructor interaction or a social robot mentor reducing interaction with human teachers).

4.4.2 Pathologization. Techno-solutionism and pathologization overlap when a given technology characterizes neurodiverse traits as deficits rather than the natural result of neurological diversity, or when interventions reinforce the double-empathy problem, placing the burden of social conformity and communication on neurodiverse individuals. Rizvi et al. found that the majority of their analyzed articles on Human-Robot Interaction (HRI) pathologized the communication behaviors of autistic individuals [49]. By examining the utilization of robots in autism, they reported that robots were employed for diagnosis, treatment, or therapy, which prompted them to classify these articles as following a medical model. Their analysis revealed that over 90% of HRI articles adhered to the medical model of disability. Our examination revealed similar findings. For instance, 43.34% of reviewed articles focused on developing intervention technology, 12.2% developed diagnostic tools and models, and 5.07% focused on training, totalling 60.61% of articles that could potentially be classified as utilizing a medical model, adopting a pathological lens of autism.

4.4.3 Essentialism. Tech “fixes” and essentialism align as they tend to oversimplify the complexity of a social problem—essentialist beliefs ignore diversity and individual autonomy within heterogeneous social groups, while techno-solutionism inadvertently reproduces social inequities by overlooking crucial questions about the root causes of the problem [50]. Rizvi et al. examined the demographic information (e.g., age and gender) of the participants and reported that these studies perpetuated “the historical masculinization and infantilization” of

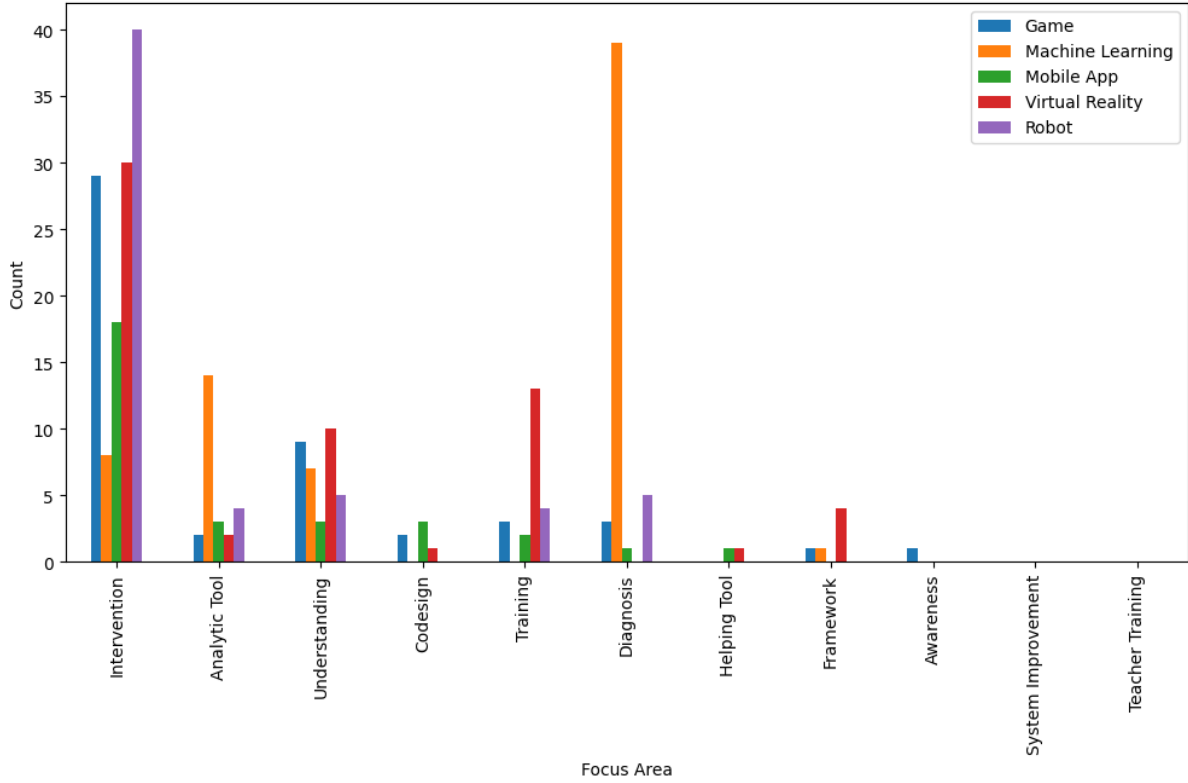


Fig. 15. Co-occurrence between technology and areas addressed.

autism [49]. Our analysis shows a similar trend. For instance, we found that 60.79% of the analyzed articles did not report gender information, and another 32.27% had a male-dominant gender representation, indicating that autism computing research contributed to the historical masculinization of autism. Additionally, our analysis revealed that only 10.13% of studies focused on autistic adults, and the majority concentrated on children, adolescents, and young adults. Our data strengthens Rizvi et al. [49] and Sharmin et al.'s [51] claim that, despite autism being a lifelong condition, autism computing research leans on children, perpetuating infantilization of autism.

4.4.4 Power Imbalance. Techno-solutionism overlaps with power imbalances when non-autistic researchers design tech “fixes” for autistic people without involving them in the process. Such solutions are intended to be helpful yet may involve benevolent paternalism—often ignoring lived experiences, risking producing solutions that are inaccessible, too expensive, or detached from real needs. Rizvi et al. reported that almost 90% of their reviewed articles did not incorporate input from the autistic individuals during the design process [49]. They argued that excluding autistic individuals introduces a power imbalance since the research design and objectives were shaped by the researchers, therapists, and special education teachers, and not by the autistic individuals. Our analysis revealed that only 6.75% of articles involved a co-design process, further highlighting this power imbalance present in autism computing research. In addition, our analysis revealed that 77.3% of the participants were children or adolescents under the age of 18. As the articles did not mention seeking consent from these participants, it is unclear how much control the participants had over the research and design process.

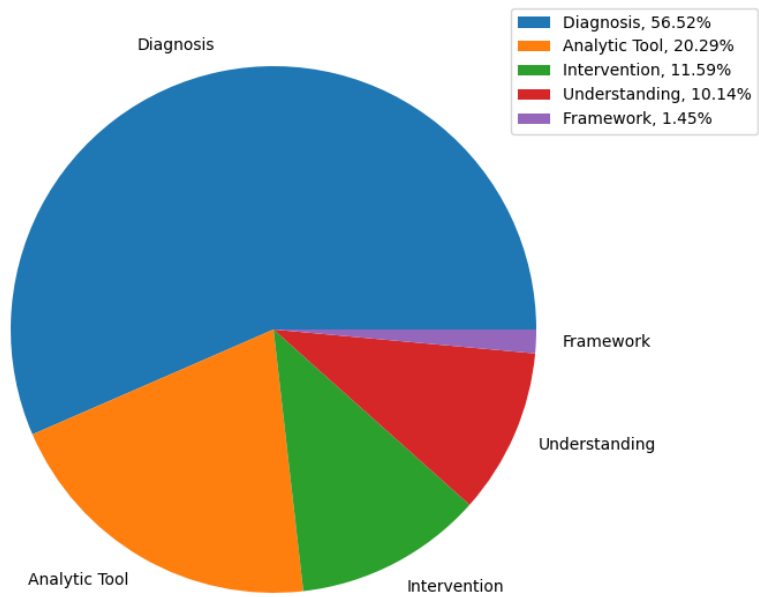


Fig. 16. Distribution of machine learning models

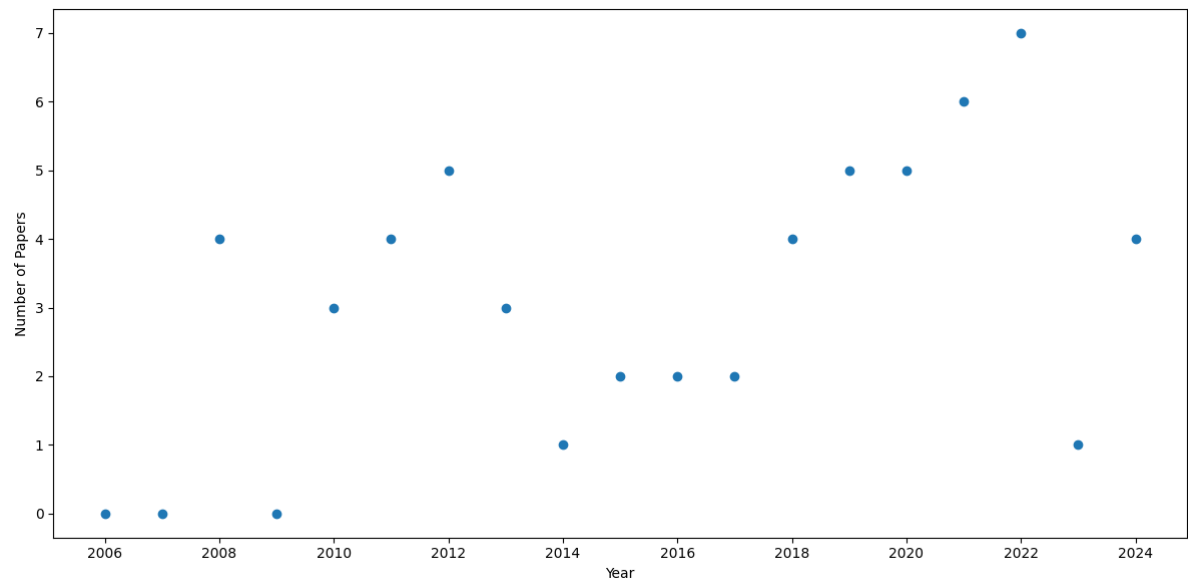


Fig. 17. Timeline of robotics usage

Furthermore, 72.41% of articles were authored by researchers affiliated with institutions in five developed countries: the USA, the UK, China, Italy, and Australia, and none were written by researchers affiliated with any African country. This signifies a potential power imbalance, as findings reported by these articles predominantly reflect the perspectives and practices of the developed countries.

5 DISCUSSION AND DESIGN IMPLICATIONS

5.1 Inclusiveness Is Imperative in Autism Computing Research

Our findings reveal that computing research on autism lacks inclusiveness in several demographic dimensions. In this section, we will discuss the resulting implications and suggest guidelines where applicable.

5.1.1 Gender Imbalance. In recent years, data-driven research has seen tremendous growth, especially in artificial intelligence and machine learning, with a strong focus on creating diagnostic tools and predictive models. The effectiveness and robustness of these tools depend on the underlying dataset, and the use of imbalanced datasets negatively impacts their performance. Among others, demographic dimensions such as gender identity are critical in creating robust models that can diagnose autism. As mentioned in sections 4.1.1 and 4.4.3, a significant percentage of reviewed articles (60.79%, 324 articles out of 533) did not report gender information, one of the most critical demographic dimensions. Also, males are significantly over-represented, while non-binary participants and their perspectives are mostly missing. This lack of inclusion is important to address, as research indicates that non-binary individuals are more likely to be on the spectrum than gender-binary people [20]. However, we have observed an increase in demographic details for these groups over time.

In addition, several articles reported intentionally recruiting only male participants to eliminate gender influence. Also, several others mentioned being unable to recruit female participants, citing the great gender disparity [34, 35], a fact that is now being questioned as a result of potential misdiagnosis of autistic females [20, 27]. If this gender gap is a result of a lack of understanding of the presentation of autism in autistic females [20, 27], intentionally excluding female participants will exacerbate misdiagnosis. The gender imbalance in autism computing research is the rule, and not the exception, when compared to other research domains [32], which has critical consequences.

Lack of gender inclusion can lead to biased and skewed technology. For example, most analytical tools utilize machine learning to develop diagnostic (e.g., autism detection using facial expression) and predictive models (e.g., stress prediction), whose effectiveness and robustness depend on the quality of underlying data. If the data is biased, the models will be biased, resulting in subpar technology. Also, gender misrepresentation can lead to technology design (e.g., games, mobile apps) that may feel exclusionary to many users who reported experiencing toxic behavior and negativity, limiting their use [31, 46]. Using gender-inclusion guidelines while designing technology may enhance participation, as users with varied gender identities will perceive them as safe. As autistic individuals report higher rates of gender deviancy from their gender assigned at birth [61], it is imperative that researchers pay attention to gender-inclusive design guidelines.

5.1.2 Imbalance of Age Distribution. Our findings reveal that a significant portion of autism computing research is focused on autistic children and adolescents, which also influences the types of technology designed by the research community, reported in sections 4.3 and 4.4.3. Similar findings were also reported in [51]. Our search didn't find any research focused on the autistic elderly or their technology needs — indicating a critical gap that needs to be filled. While there is research focusing on autistic young adults and adults, they are underrepresented. This issue is not limited to computing research, but instead reinforces interdisciplinary findings that older autistic adults face unique challenges and are underrepresented in research [43]. The recent increase in autism computing research includes many articles that focus on early diagnosis, assisting in social communication, and behavioral issues, all targeted towards children. This narrowed focus ignores the needs of young adults, adults, and the

elderly. Consequently, very little research focuses on topics such as independent living, employment search and retention, long-term relationship development, and maintenance—topics that are critical to the quality of life of autistic individuals. As autism is a lifelong condition and each phase of life introduces new challenges to autistic individuals, researchers must pay attention to the design and development of supportive technologies for autistic individuals of all ages. In addition, research in genetics and neuro-development suggests that autism may be caused/influenced by genetics and environmental factors [2, 28], and understanding the causes will necessitate a deeper understanding of not only data collected from autistic children but also their parents and environment.

5.1.3 Imbalance of Geographical Location of Research. As mentioned in section 4.1.3, our findings reveal that autism computing research is highly concentrated in economically developed, Western continents and countries. Of the top 10 research-producing countries, only two are from Asia (China and Japan), and none are from Africa (see Figure 7). However, we acknowledge that other publication venues likely reflect different geographical distributions that were not captured in our dataset. It is well-acknowledged that not only is technology design influenced by culture, but culture also greatly influences the acceptance and adoption of technology [15, 56]. Culture is intertwined with users' geographical location and should be carefully considered while designing technology. Autism is believed to be caused by genetics and environmental factors, and as such, any autism support technology needs to be culturally appropriate. For example, research aimed at behavioral intervention needs to consider what would be culturally appropriate behavior. For instance, Western culture encourages direct eye contact, while Eastern culture often views it as a sign of disrespect. Ignoring these differences may result in poor effectiveness of the designed technology. As many autism support technologies focus on assisting users in learning to engage appropriately in social situations, these technologies should be designed using a culturally sensitive lens.

5.1.4 Cultural and Linguistic Diversity Shapes Scientific Terminology. The language use we analyzed—namely, between person-first (e.g., “individuals with autism”) and identity-first (e.g., “autistic individuals”) terminology—reflects cultural preferences that are prevalent in Western, English-speaking research communities.

The language choices in the reviewed articles are embedded in cultural and linguistic contexts, and terminology preferences can vary significantly across global regions. For example, in some non-English-speaking countries, it is possible that no linguistic equivalent for person-first language exists, or such phrasing may carry different connotations. In many cultures, collective identity is emphasized over individual identity, which can influence how disability or neurodivergence is described. Additionally, some communities may prioritize medical or diagnostic terminology over neurodiversity-affirming language due to sociocultural or policy-driven norms.

If we collected research articles from a different publication platform, our findings could potentially reveal different terminology trends and framing practices. For instance, language influenced by local translations of the DSM or ICD (International Classification of Diseases), or by government policy regarding disability, may shape both the adoption of terminology and the underlying assumptions in research. Therefore, while our current dataset provides an important view of terminology trends within ACM, we recognize that these trends may not generalize globally. In addition, our research limitation highlights the importance of future research investigating language-preferences and cultural norms that can shed light on research published outside Western, English-language platforms.

5.1.5 Gender Identities, Game Design, and Gaming Culture. We found that games are one of the most researched technologies in autism computing research. Gamification serves as a way to naturally increase engagement with interventions, making it a popular choice for technology for autistic children. Furthermore, several reviewed articles discussed online autism groups centered around video games. These groups offer a safe community of like-minded people who can provide support to each other. The effectiveness of these groups relies on their ability to maintain an inclusive and non-judgmental environment. However, historical male-dominant attitudes in the

gaming space, combined with the skewed autism diagnosis gender ratio, may contribute to autistic women/girls feeling less comfortable with traditional game design and culture. Therefore, game developers and gaming community members should strive to model inclusive and anti-sexist behavior. Moreover, when examining the use of assistive technologies in health and education, some researchers, such as [55], question whether joy, relaxation, and fun are truly centered in games built for autistic children.

5.2 The Importance of Adopting an Inclusive, Strength-based Mindset

5.2.1 *Strength-based and Unique Technologies.* Computing research on autism primarily focused on addressing perceived flaws in autistic individuals and follows a deficit-based approach. Such an approach can lead to assumptions about the capabilities of a marginalized group, preventing researchers from thinking outside the box for solutions [57] and may encourage utilizing a deficit-based framework, which frames autism as a collection of deficits that should be corrected while ignoring autistic strengths, perpetuating ableist attitudes.

At present, technology designed with autistic strengths in mind is underrepresented, albeit such research may be more accepted by autistic individuals. In addition, the majority of computing research did not address topics important to autistic individuals, such as aging, parenthood, sexuality, intimacy, amplifying autistic voices, cultural and economic sensitivity, and employment [38]. We encourage autism researchers to broaden the scope of autism research by exploring avenues that are critical for the empowerment and well-being of autistic individuals.

5.2.2 *The Importance of Using Inclusive and Respectful Language and Tone.* Language and tone are powerful and can impact the target users, research directions, and society [7]. This is especially important for autism computing research, considering the stigma associated with autism in society. For example, terms such as disease, sickness, and illness to describe autism indicate that autistic people need to be ‘fixed’ by removing their ‘autistic traits,’ a sentiment that many adult autistic individuals, autistic advocates, and allies disagree with. Such research often proposes technologies that aim to ‘fix/cure’ autistic individuals by changing behavioral traits (e.g., fixing eye gaze, reducing instances of stimming) that are often beneficial for them. On the other hand, terms such as condition, characteristics, etc., portray autism as part of an individual’s identity. Many autistic individuals, advocates, and allies prefer this later framing as it focuses on designing solutions to accommodate their unique needs. In addition, some research uses language and tone that can be considered dehumanizing. For example, comparing autistic behaviors with robots and virtual agents, attributing challenges in social communication as a characteristic associated with machines [14, 24] is unethical, ableist, and discriminatory [7, 53]. While our findings reveal that autism computing researchers are slowly shifting towards the use of more appropriate language and tone (such as using identity-first language and aligning with the social model of disability), the research community needs to become more mindful regarding the use of language and tone. We believe that area-specific knowledge, especially when working with underrepresented, marginalized, or special user groups, is critical for research effectiveness, also suggested by [37].

5.3 The Need for Better Alignment of Target Users’ Needs and Designed Technology

Machine learning, robotics, games, virtual reality, and mobile apps were the most prevalent technologies utilized as an autism support system. While these systems were generally reported to have positive effects on social and cognitive learning outcomes, the use of varying evaluation frameworks limited their generalizability [10]. Serious games, often geared toward autistic children, have shown promise in social learning, which would benefit from utilizing standardized evaluation frameworks, robust methodology such as randomized control trials (RCTs), and collaborations between technology designers and psychology experts [9, 21]. Furthermore, Azadboni et al. found that serious games that utilize multiple modalities (e.g., audio and visual, or mental and physical tasks) were less effective than those using only one modality, suggesting that effective game design would benefit from

psychological expertise and an understanding of how factors like masking influence engagement in autistic individuals [58].

In the Human-Robot Interaction (HRI) domain, researchers suggest robots may support some autistic individuals' preferences for predictability — allowing a scaffolded increase in social complexity and facilitating interactions with human caretakers [40]. VR effectiveness was influenced by age, session length, and simulation style [12]. Mobile apps were more effective when integrating a realistic style (as opposed to an animated style) and everyday activities [30]. Machine learning represented 10.57% of developed technologies in our dataset and was used for diagnosis approximately 60.7% of the time. While machine learning diagnosis may offer quicker, more affordable, and objective results, the clinical adoption and efficacy of such models may depend on whether they rely on behavioral assessments, neuro-imaging, eye-tracking, or other data, since some approaches align more closely with DSM-5. In addition, many of these research articles do not report their effectiveness in real-life diagnosis [13]. These findings indicate that autism support technology must be evaluated for pre- and post-interaction effects, involve autistic individuals, caregivers, and practitioners in the design, development, and evaluation process - utilizing co-design practices.

While these technologies offer opportunities, researchers should carefully consider several aspects while envisioning technology. For example, social skills interventions may align with neurotypical expectations of communication by disregarding autistic experiences and amplify the double empathy problem [23]. In addition, recent literature has criticized the role of HRI research in amplifying harmful narratives — using robots to model social interactions may imply a sub-human status of autistic individuals or aid stereotypes of autistic individuals as 'robotic' [49]. Such technology can also potentially reinforce the normativity of neurotypical methods of communication.

5.3.1 Control, Surveillance, Privacy, and Monitoring. Use of smart technologies in autism computing research is rapidly increasing, but privacy protections for these technologies are underrepresented in autism computing research. 246 papers (46.15% of our dataset) included technologies with clear privacy implications — such as AI model transparency, biometric and physiological data collection, and location or behavioral tracking. As we only captured the primary research focus of each paper, privacy was not identified as a central topic of investigation throughout our dataset. While some papers addressed privacy-related themes such as safety, marginalization, and empowerment, they only account for 6.56% of our dataset.

Data collection technology for autistic children, who may be particularly vulnerable from an autonomy perspective, should also be explored from privacy and consent perspectives, and appropriate safeguards should be implemented. Autistic children monitored by technology may be subject to more surveillance than their neurotypical peers, possibly incurring additional cognitive load [62]. In addition, wrist-worn sensors can require a desensitization period, manifesting aesthetic, social, and sensory challenges for autistic individuals [5]. The benefits of data collection must be weighed against the discomfort of wearable technologies for autistic users.

5.3.2 Screen Time and Technology. Yuan et al. suggested, though with low confidence, that an ASD diagnosis correlates with higher overall screen time and an increased risk of screen addiction [65]. This finding raises questions about the increasing use of serious games and mobile apps that are targeted towards autistic individuals, a critical aspect that technology designers should consider. Additionally, using many technological interventions may serve to motivate autistic individuals, especially children, to spend more time online, which may go against lower screen-time goals. It is also important to consider whether the increasing number of technological solutions is developed because of autistic individuals' natural affinity for technology [22].

5.3.3 Financial Accessibility. The availability and cost associated with different types of autism support technology (e.g., robots, smartphones, mobile-based gaming, and communication support systems) make them inaccessible

to many autistic individuals. For example, the predominant use of robotics for aiding children in social communication is problematic due to the high cost associated with robots. To reduce financial burden, researchers should investigate cost-effective technological solutions as autism is an expensive condition to diagnose and manage [17, 26], and 95% of autistic individuals live in low- and middle-income communities and countries [29].

5.3.4 Need for Involving Autistic Individuals in the Design Process. Autism support technology should be driven by the needs and experiences of autistic individuals [11]. Since many autistic individuals experience challenges in effectively communicating their needs, other stakeholders such as caregivers, practitioners, and autism experts should be included in the design process. In addition, as autism is a spectrum and different autistic individuals have different characteristics and needs, including autistic individuals with different characteristics during the design process is critical. For example, our research reveals that non-speaking and ASD level 3 autistic individuals are underrepresented in autism computing research. To utilize co-design, researchers must include the voices of autistic individuals from all levels and all characteristics. Another relevant consideration is the technology-savviness of autistic individuals. While we didn't examine technology-savviness as one of the characteristics, we recommend that researchers consider technology-savviness along with technology-experience because autistic individuals may have varying competencies when utilizing technology. Furthermore, technology that tries to bridge the gap between autistic and allistic communication often places the burden of change on autistic individuals, assuming they alone are responsible for closing that gap [23]. For example, many communication and interaction support technologies focus on helping autistic individuals to communicate, interact, or express emotions by creating training technologies focused on teaching allistic ways of communication, interaction, or expression. However, technology focused on teaching allistic individuals how to communicate, interact, or understand autistic expressions is scant.

By investigating the disconnect between common goals and language of the #ActuallyAutistic community on Twitter and those of researchers, Guberman suggested that autism-computing research should seek to build collaborative relationships with online autistic communities [18]. Their recommendations involved eliminating paywalls for autism computing research articles, funding autistic individuals to attend academic conferences, and obtaining consent and crediting social media users when collecting datasets. We agree with their sentiment that researchers should take initiative to engage with online autistic communities to bridge the communication gap while respecting community guidelines. As autistic individuals and their experiences and needs are highly diverse, researchers should be more open to reporting findings that may lack generalizability and carefully consider this variation while designing technology.

5.3.5 Relevance of IQ. A lack of awareness regarding autism and autistic traits can cause misunderstanding. For example, many articles in our repository used or reported IQ scores to discuss autistic individuals. There is no consensus on mapping IQ to DSM-5's three ASD levels. Also, existing IQ scales are based on allistic-focused datasets, which may not accurately measure the IQ of autistic individuals [25, 63]. A lack of understanding regarding autism and its relationship to IQ resulted in different articles using different thresholds to classify autistic participants, which raises questions regarding the validity of their findings.

5.4 Alignment with HCI and IMWUT Research Community

Recent years have seen tremendous growth in autism research in computing. A vast majority of this research focuses on designing technological solutions that aim to address critical aspects such as autism diagnosis, designing and delivering behavioral interventions, and educating and training autistic individuals for different aspects of life. The increasing popularity of emerging and smart technologies such as AI-based systems, robotics, wearable sensors, machine learning, and deep learning in developing autism support systems signifies a critical need for a careful examination of their effectiveness in aiding autistic individuals. While such attention promises

a better technology usage experience for autistic individuals, we learned that there is a disconnect between the needs of autistic individuals and the developed technology. Also, some areas of autism research received more attention (e.g., diagnosis, technology to aid social interaction) while others received minimal attention (e.g., independent living, employment seeking and retention, relationships). This mismatch may be attributed to a considerable lack of use of co-design and human-centric research methods. Our research takes a first step in providing a comprehensive understanding of the current state of computing research in autism, highlighting areas that require more attention and pointing out disparities (such as a lack of inclusiveness in terms of gender, age, and race) that limit the effectiveness of autism research. As smart and emerging technologies become common as autism support technology, we believe researchers envisioning such technologies will benefit from a better understanding of the landscape of autism research, especially pertaining to computing technology. We offer actionable guidelines that HCI and technology researchers, who are integral to the IMWUT research community, can use to design more effective and inclusive autism support technologies.

6 LIMITATIONS AND FUTURE RESEARCH DIRECTIONS

In this section, we discuss the limitations of our research and plans for future direction.

Data Corpus: Our research goal is to offer a detailed understanding of the landscape of autism research in computing based on the ACM Digital Library and identify under-represented areas that deserve more attention. Instead of collecting articles from all databases (e.g., ACM, IEEE, PubMed), as a first step, we only focused on the ACM Digital Library, which may potentially exclude technology-focused articles published in IEEE and other non-computing domains (e.g., PubMed). Additionally, we excluded non-English papers. Also, one of our search keywords, ASD, appears alongside many frequently coexisting conditions (e.g., ADHD, cognitive impairments, anxiety, and epilepsy). As we specifically searched for articles that primarily focused on autism, articles that primarily focused on coexisting conditions such as ADHD were excluded. We also did not include "neurodiversity" as a keyword to limit the search to articles primarily focusing on autism, an intentional design decision stemming from our assumption that articles with this focus would include autism along with other terms and would be included in our search results. While including publications from IEEE or other platforms might improve our findings and offer additional perspectives, we believe that our analysis of a large dataset (N=533) from the ACM Digital Library is an important first step and will significantly enhance our current understanding of autism computing research.

Labeling country of origin: To mark the geographical location of the research, we used the researcher affiliations (location of the institution the researchers worked at/published for). However, this might not reflect where the studies themselves were conducted. A significant portion of our analyzed articles recruited participants, and their culture might be different from that of the research team members, which we derived from their affiliation. This points out an exciting area of future research that can shed light on whether and how researchers' cultural identity influences their research directions, choice of technology, and language and tone used.

Data categorization and visualizations: To enhance the intuitiveness of visualizations, we grouped our data into meaningful categories. However, the creation of these categories was challenging. First, some data points fell into multiple categories (e.g., technology types), and second, there were no directions for categorizing the data due to a lack of relevant details (e.g., participant types). For example, figures representing technology needed grouping as there are more than a hundred different types of autism support technology. While some technologies belong to only one category (machine learning diagnostic tools), others can be categorized into multiple groups (gamified mobile applications can be categorized as games or mobile applications). To avoid over-representation, we grouped them into one specific category (i.e., mobile application).

Data for which enough details were absent presented additional challenges during grouping. For example, we used 17 different codes to describe autistic participants (e.g., level 1, high functioning, Asperger's, moderate).

This is partially caused by the changing characterization of autism by the official diagnostic manuals (DSMs). For example, before DSM-5, which was published in 2013, autism was categorized based on functional support needs (e.g., high functioning, low functioning, etc.), and Asperger's was not considered an autism category. Many articles published before 2013 utilized these categories. DSM-5 introduced new categories such as Level 1, Level 2, etc., and merged Asperger's diagnoses with autism. Many articles published after 2013 utilized these new level-based categories to describe their participants. As no precise mapping exists between the functionality-based and level-based categories, it is impossible to connect these two categories. In addition, many articles used mild, moderate, and severe to describe their participants based on their symptom intensity, and it is unclear how these categories map to either the functionality or the level-based categories. We chose to group the participant data using their different categories and not connect them to avoid misrepresenting participants.

In addition, our corpus contained only English-language articles, which predominantly featured articles from the USA and UK. As such, our findings regarding terms used may not be generalizable for research articles written in other languages that use a different categorization.

Subjectivity: Researchers' positionality and motivations, such as the use of the medical vs. social model or the identification of negative language, were infeasible to code due to the lack of explicit mention of the models and the subjective nature of this categorization. As mentioned earlier, some code groups presented significant overlap. Primarily, the technology, problem, and challenge code groups included a degree of subjective categorization. We identified 130 unique technology codes, later grouping these instances into 11 umbrella themes. Our finalized categories also overlap, reflecting technology's multi-faceted purpose and design. For example, communication devices (AAC, social scripts, tutor) shared inherent similarities with devices (tablets, touchpads, laptops) and applications (games, social media, personal assistant). Additionally, we chose to primarily view trends in terms of the count of articles fulfilling certain criteria. This will be useful for understanding what researchers are choosing to publish about, but needs to be used in conjunction with articles that specifically evaluate the efficiency of individual technologies.

As our next step, we plan to utilize natural language processing (NLP) tools to gain a deeper understanding of the language and tone used in the articles and identify patterns regarding keywords, metadata, and models of disability. We also want to expand our research by conducting similar reviews with articles published in IEEE and other research platforms.

7 CONCLUSION

The last two decades have observed a steady growth in autism research in computing, not only in terms of the number of research articles published but also in the diversity in research motivation (e.g., early detection of autism, improving quality of life), different types of technology used (e.g., sensing and tracking technologies such as smart watch and smart garments, humanoid robots, machine learning models), and the diversity of the problem areas (e.g., emotion detection, interview training) they addressed. While this growth promises tremendous potential to improve the experiences of autistic individuals and different facets of their lives, it also highlights the possibility of ill-designed or biased technology that can negatively impact autistic individuals. In addition, our findings point out that autism computing research is highly concentrated in economically developed countries and may offer knowledge that is not reflective of autistic individuals' experiences who live in underdeveloped countries and communities. Furthermore, utilization of cutting-edge technology, which often is expensive, will exacerbate this situation by increasing the digital divide between individuals from economically developed and underdeveloped communities and countries. To aid researchers and technology developers in identifying areas that deserve more attention, we analyze the landscape of autism computing research in the ACM Digital Library and offer insight into existing research, their motivations, strengths, and weaknesses. Using the social model of

disability and a strength-based approach, we advocate for inclusiveness in every aspect of autism computing research design.

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A Methodological Transparency and Reproducibility

To ensure the reproducibility of this work, a list of all 533 papers analyzed is provided as auxiliary material. The list can be accessed via this webpage: <https://neat-lab.cs.wvu.edu/IMWUT/IMWUT-ACMDL-PaperList.pdf>.